# Exercise 4
# Single-Stage Amplifiers

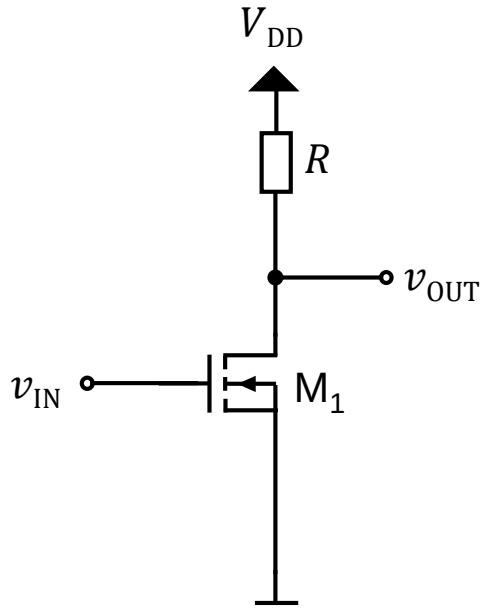
Winter Term 2025/26

Nima Shahpari, Ph.D.

- Task 1: Common Source Amplifier with Resistive Load

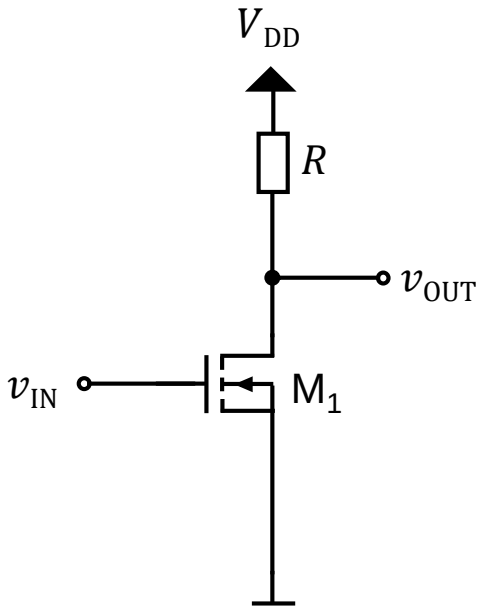
$V_{DD}$ = 1.2 V, $I_D$ = 8 µA, W/L = 2/2, $R$ = 0.1 MΩ

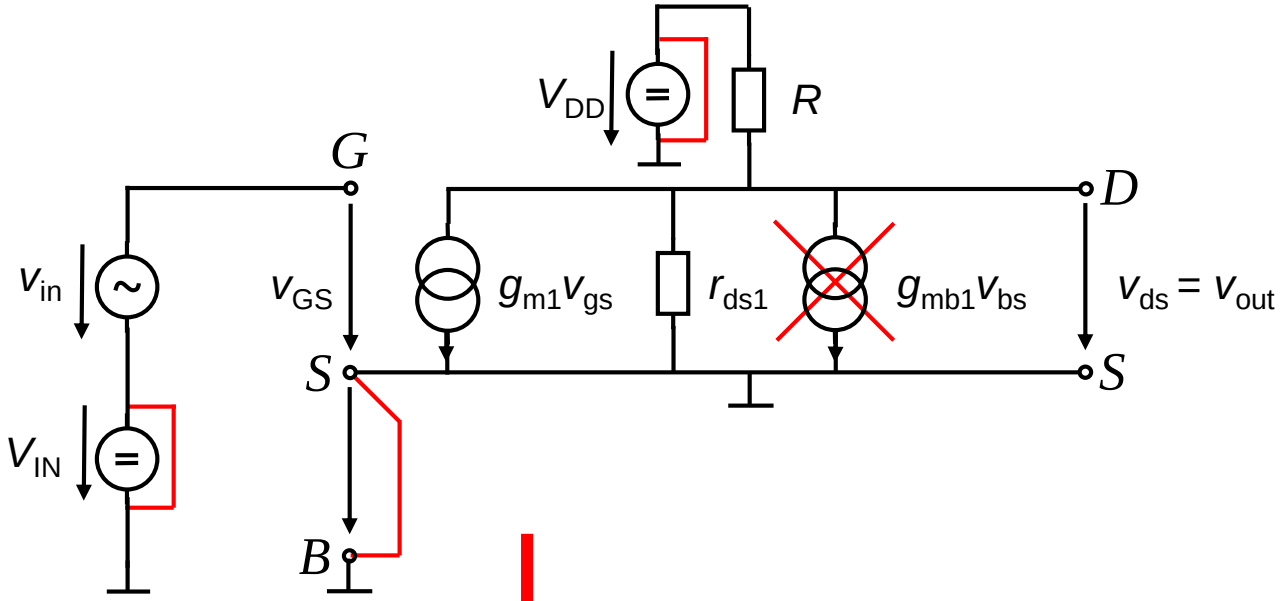
$\beta_0$ = 360 µA/V$^2$, $V_{th0}$ = 0.19 V, $\lambda$ = 0.088 V$^{-1}$

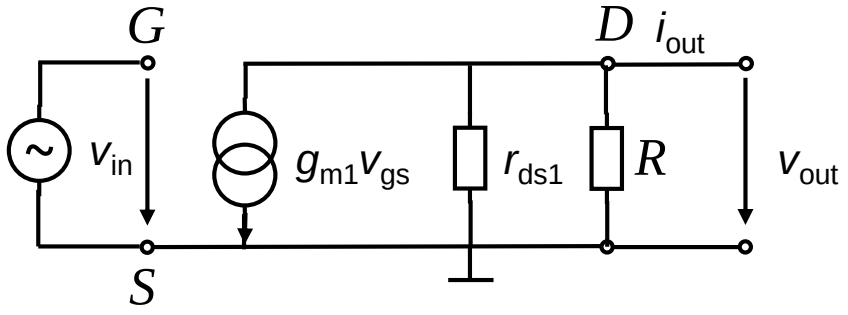


# Exercise 4 – Task 1

- 1.1. Draw the small-signal equivalent circuit and derive an equation for the voltage gain!
  - a.1. Calculate the widths of $T_R$ and $T_1$. Channel-length modulation is to be considered.

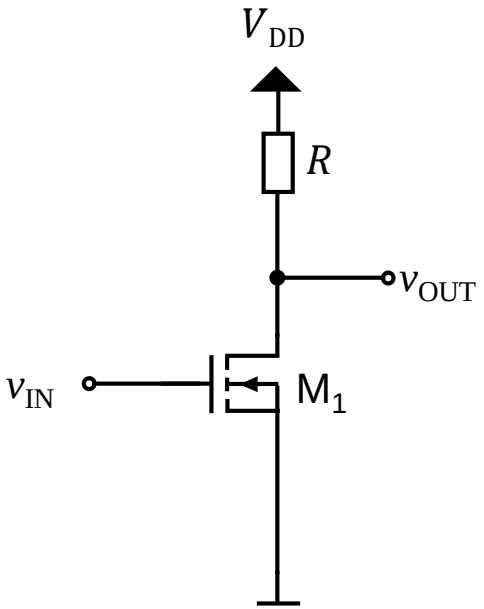


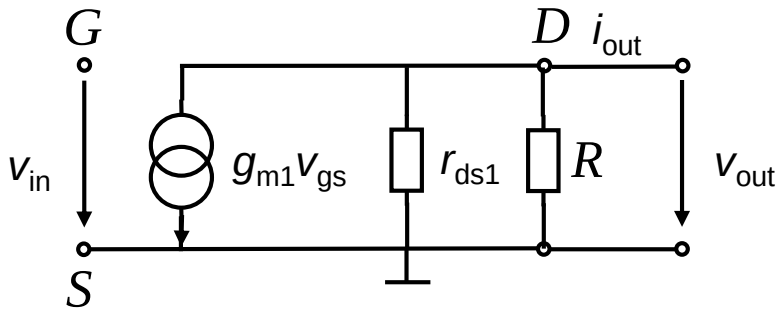



# Exercise 4 – Task 1

- 1.1. Draw the small-signal equivalent circuit and derive an equation for the voltage gain!
  - i.e., $i_{\mathrm{OUT}} = 2i_{\mathrm{REF}}$. Calculate the width of $\mathrm{T}_2$ accordingly.



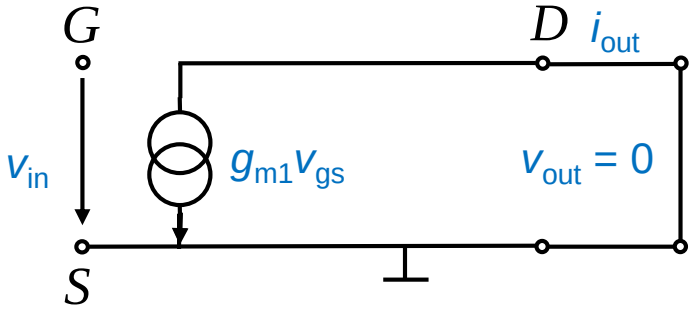

voltage gain: $A_{\mathrm{v}} = -G_{\mathrm{m}}r_{\mathrm{out}}$

# Exercise 4 – Task 1

- 1.1. Draw the small-signal equivalent circuit and derive an equation for the voltage gain!

$$A_V = -G_\mathrm{m} r_\mathrm{out}$$

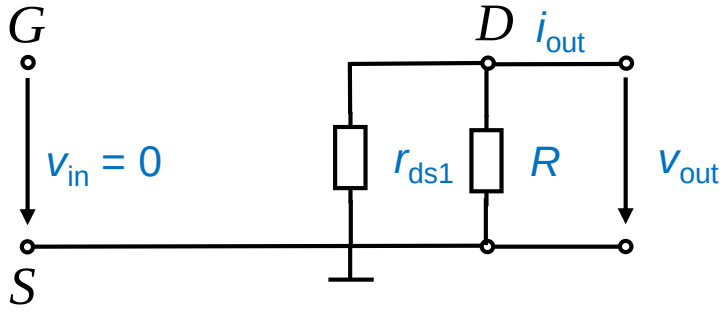
short-circuit current




$$G_\mathrm{m} = \left.\frac{i_\mathrm{out}}{v_\mathrm{in}}\right|_{v_\mathrm{out} = 0\ \mathrm{V}}$$

$$i_\mathrm{out} = g_\mathrm{m1} v_\mathrm{in}$$
$$G_\mathrm{m} = g_\mathrm{m1}$$

open-circuit voltage



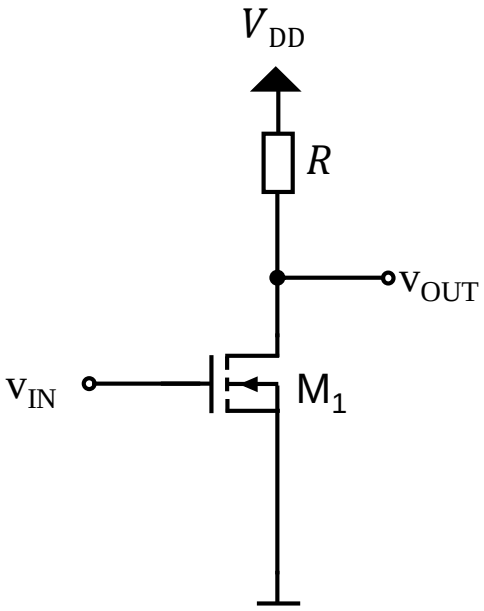

$$r_\mathrm{out} = \left.\frac{v_\mathrm{out}}{i_\mathrm{out}}\right|_{v_\mathrm{in} = 0\ \mathrm{V}}$$

$$v_\mathrm{out} = i_\mathrm{out}(R||r_\mathrm{ds1})$$
$$r_\mathrm{out} = (R||r_\mathrm{ds1}) = \frac{R r_\mathrm{ds1}}{R + r_\mathrm{ds1}}$$

voltage gain: $A_\mathrm{V} = -g_\mathrm{m1}(R||r_\mathrm{ds1})$

# Exercise 4 – Task 1

- 1.2 Assuming that M1 operates in saturation, calculate the small-signal gain!



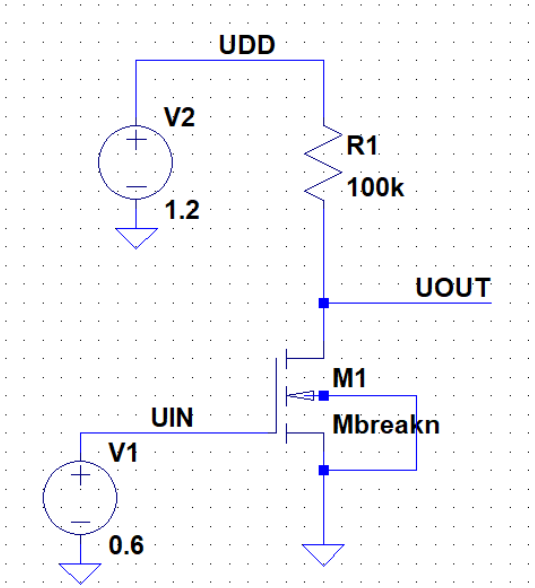

$$A_V = -g_{m1}(R||r_{ds1})$$

$$A_V = -\underbrace{\sqrt{2\beta_0 I_D \left(\frac{W}{L}\right)}}_{75.9\,\frac{\mu A}{V}} * \underbrace{\left(\frac{1}{\underbrace{\lambda I_D}_{14.2M\Omega}} \middle\| R\right)}_{99.3\,k\Omega}$$

$$A_v \approx -7.5$$

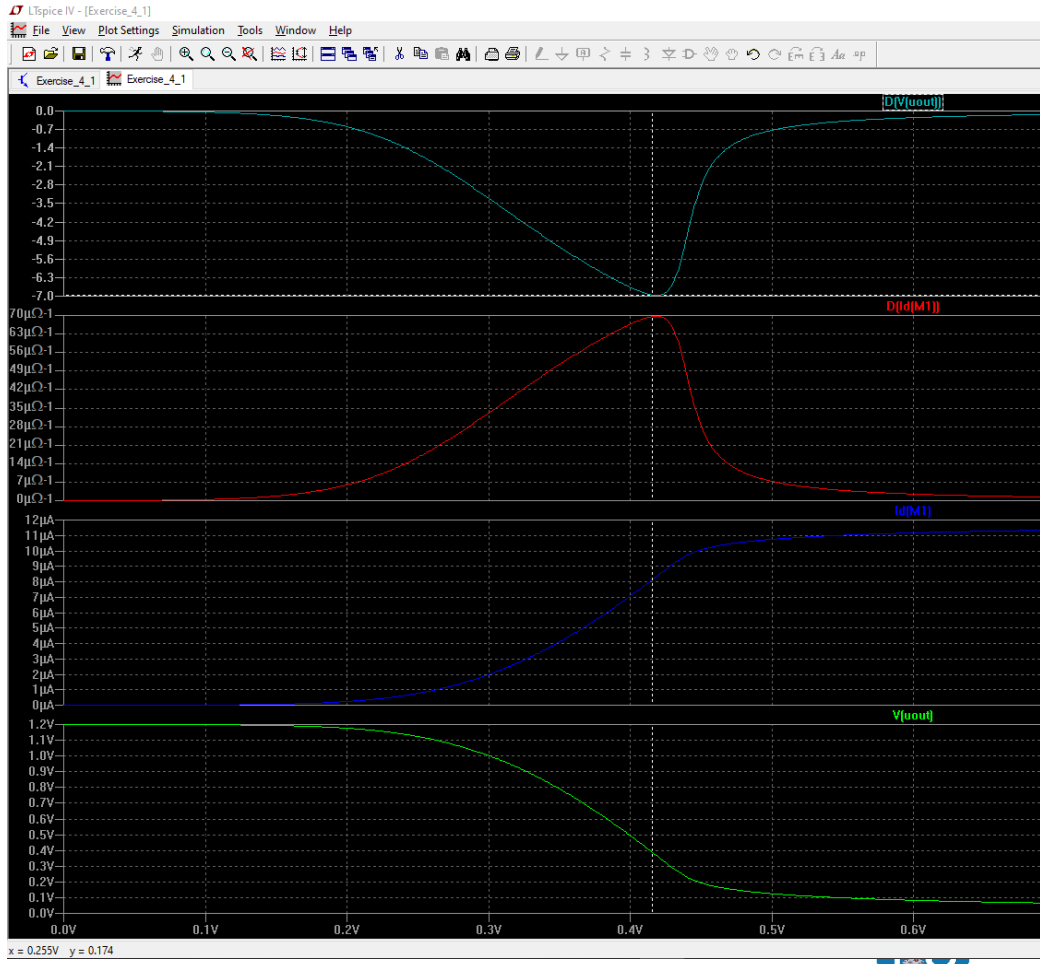
# Exercise 4 – Task 1

- 1.3 Simulate the small-signal gain in LTspice and compare the result to the calculated value!




**Hint: Perform the derivation of $v_{OUT}$ with respect to $v_{IN}$.**

**Perform a DC-sweep over the voltage $v_{IN}$ and plot $v_{OUT}$ versus $i_D$. From the menu bar of the result window, select *Plot Settings* → *Add trace*. In the pop-window, type in the derivative operator *D( )* and include the output voltage from the available data. Read out $D(v_{OUT})$ at $I_D$ = 8 µA.**

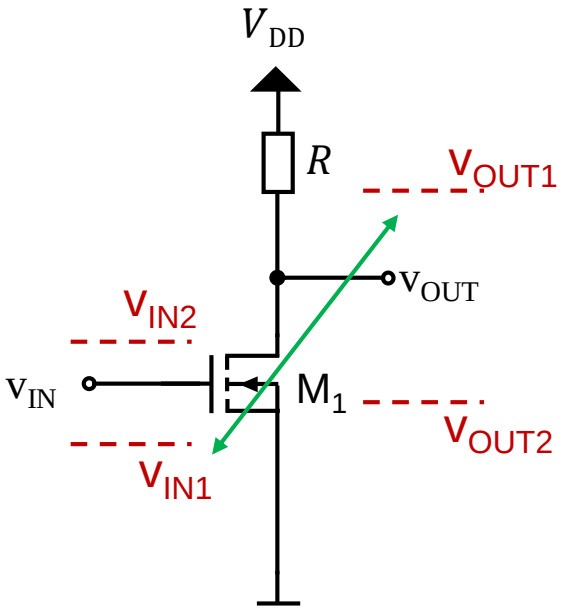



| parameter | calculation | simulation |
|---|---|---|
| $g_m$ | 76 µS | 69.7 µS |
| $A_V$ | -7.5 | -6.96 |

# Exercise 4 – Task 1

1.4. The output common-mode range of the amplifier is 0.4 V to 1 V

1.4.1. Calculate the corresponding input common-mode range




1) derive an equation for $v_{OUT}$

$$v_{OUT} = V_{DD} - I_D R = V_{DD} - \frac{\beta_0}{2}\frac{W}{L}(v_{IN} - V_{th})^2 R$$

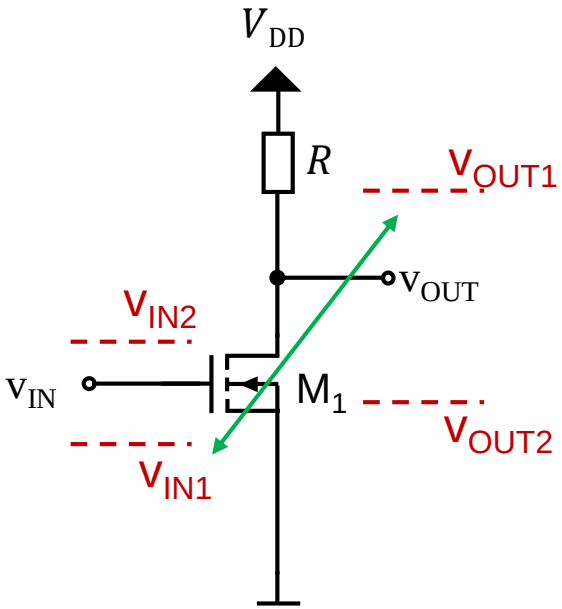
2) solve for $v_{IN}$

$$v_{IN} = \sqrt{\frac{V_{DD} - v_{OUT}}{\frac{\beta_0}{2}\frac{W}{L}R}} + V_{th}$$

# Exercise 4 – Task 1

1.4. The output common-mode range of the amplifier is 0.4 V to 1 V

1.4.1. Calculate the corresponding input common-mode range




$$v_{IN} = \sqrt{\frac{V_{DD} - v_{OUT}}{\frac{\beta_0}{2}\frac{W}{L}R}} + V_{th}$$

3) Calculate $v_{IN}$ for $v_{OUT}$ = 0.4V & 1V

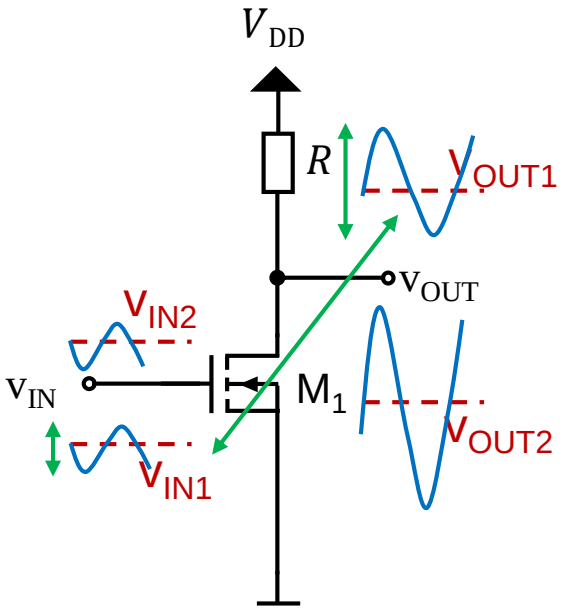
$$v_{OUT} = 1\ \text{V} \quad \Rightarrow v_{IN1} = \sqrt{\frac{0.2\ \text{V}}{180\ \mu\frac{\text{A}}{\text{V}^2} \cdot 1 \cdot 0.1\ \text{M}\Omega}} + 0.19\ \text{V} = \underline{0.3\ \text{V}}$$

$$v_{OUT} = 0.4\ \text{V} \quad \Rightarrow v_{IN2} = \sqrt{\frac{0.8\ \text{V}}{180\ \mu\frac{A}{\text{V}^2} \cdot 1 \cdot 0.1\ \text{M}\Omega}} + 0.19\ \text{V} = \underline{0.4\ \text{V}}$$

# Exercise 4 – Task 1

1.5. The output common-mode range of the amplifier is 0.4 V to 1 V

1.5.2. Calculate the corresponding small-signal gains!




$$A_v = \frac{\partial v_{\mathrm{OUT}}}{\partial v_{\mathrm{IN}}} = -G_{\mathrm{m}} r_{\mathrm{out}} = -\underbrace{\beta_0 \frac{W}{L}(v_{\mathrm{IN}} - V_{\mathrm{th}})}_{gm} \underbrace{\left(\frac{1}{\lambda I_D} \middle\| R\right)}_{r_{\mathrm{out}}}$$

small-signal gains

$$A_{\mathrm{V}}|_{V_{\mathrm{IN}}=0.3\mathrm{V}} = -\beta_0 \frac{W}{L}(V_{\mathrm{IN1}} - V_{\mathrm{th}})R$$

$$= -360\ \mu\frac{\mathrm{A}}{\mathrm{V}^2} \cdot 1 \cdot (0.3\ \mathrm{V} - 0.19\ \mathrm{V}) \cdot 0.1\ \mathrm{M\Omega} \approx \underline{-4}$$

$$A_{\mathrm{V}}|_{v_{\mathrm{IN}}=0.4\mathrm{V}} = -\beta_0 \frac{W}{L}(v_{\mathrm{IN2}} - V_{\mathrm{T}})R$$

$$= -360\ \mu\frac{\mathrm{A}}{\mathrm{V}^2} \cdot 1 \cdot (0.4\ \mathrm{V} - 0.19\ \mathrm{V}) \cdot 0.1\ \mathrm{M\Omega} \approx \underline{-7.6}$$

$A_{\mathrm{v}}$ depends on bias point!

# Exercise 4 – Task 2

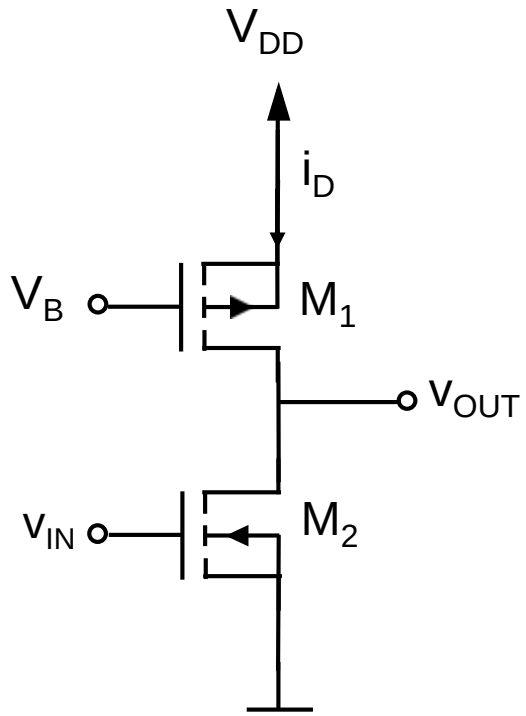
## Task 2: Common-Source Amplifier with Current Source Load

$V_{DD}$ = 1.2 V, $I_D$ = 45 µA, $v_{OUT,max} - v_{OUTmin}$ = 0.7 V, $v_{out}/v_{in}$ = -72 V/V

$\beta_{0n}$ = 360 µA/V², $V_{thn}$ = 0.193 V, $\lambda_n$ = 0.088 1/V

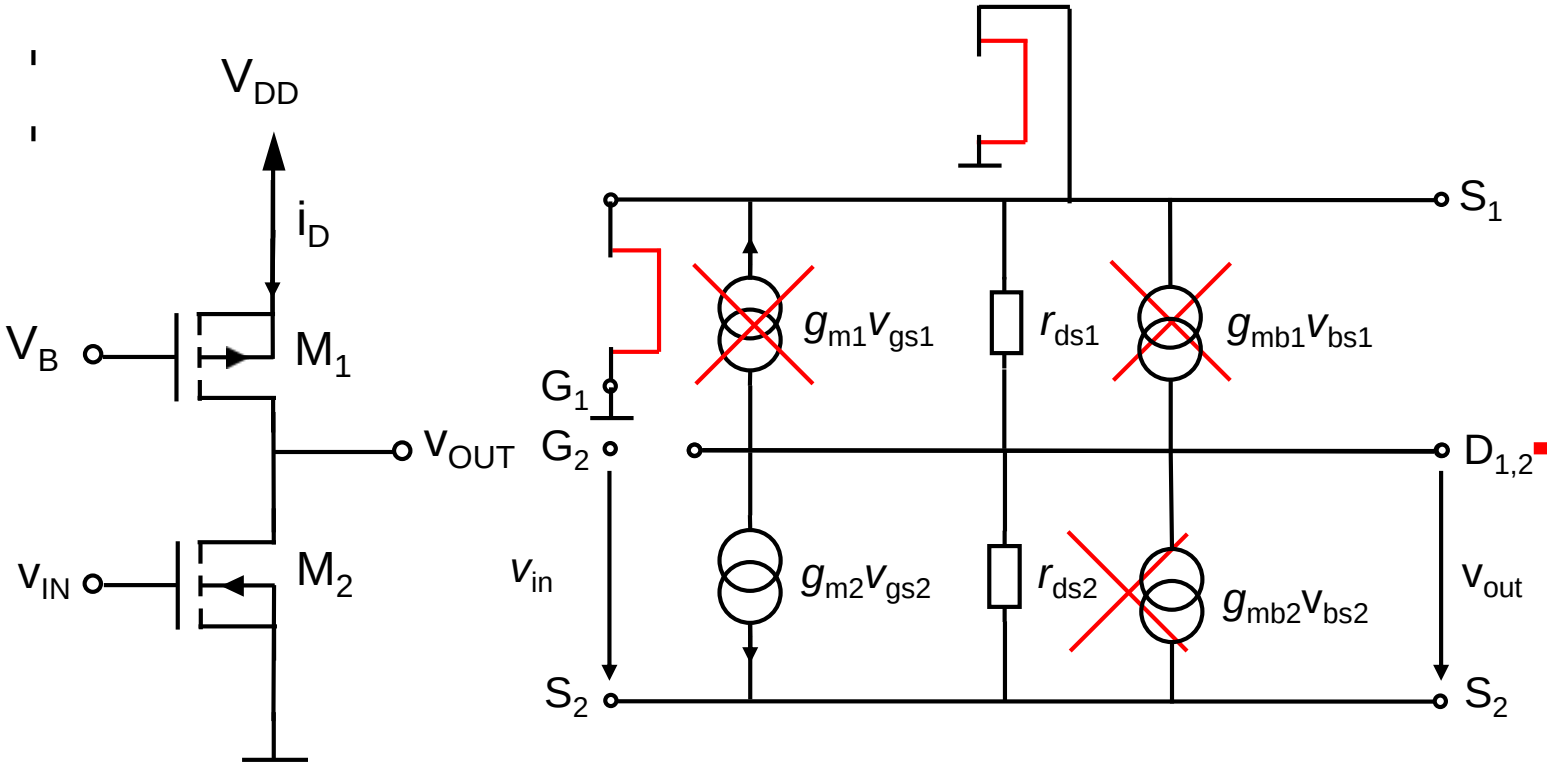
$\beta_{0p}$ = 63.5 µA/V², $V_{thp}$ = 0.215 V, $\lambda_p$ = 0.023 1/V

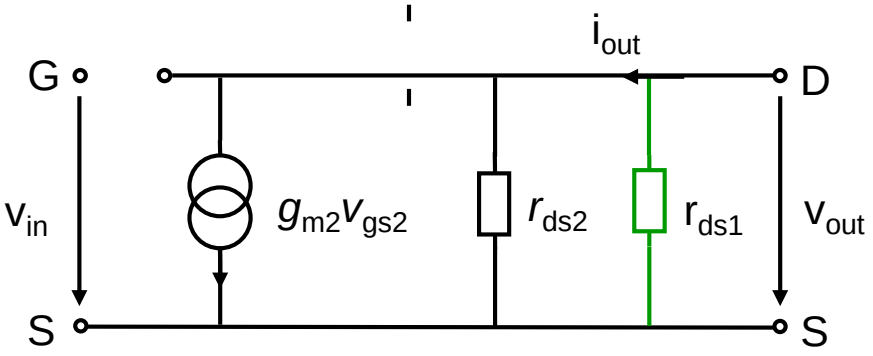
$L$ = 2 µm



# Exercise 4 – Task 2

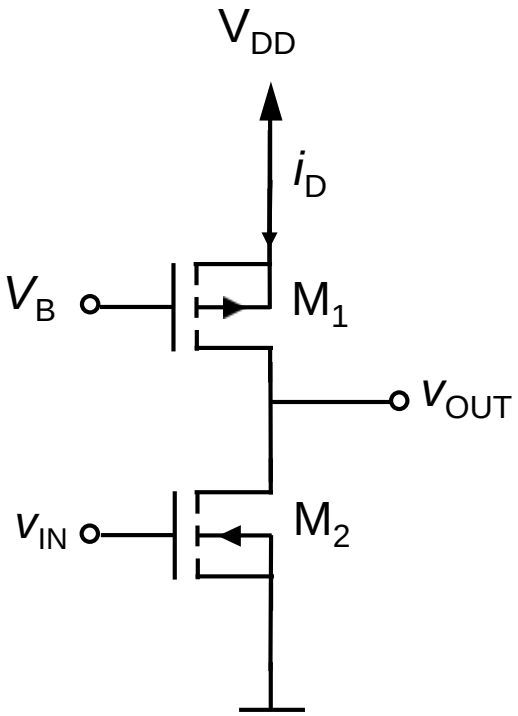
2.1 Draw the small-signal equivalent circuit and derive an equation for the small-signal gain!

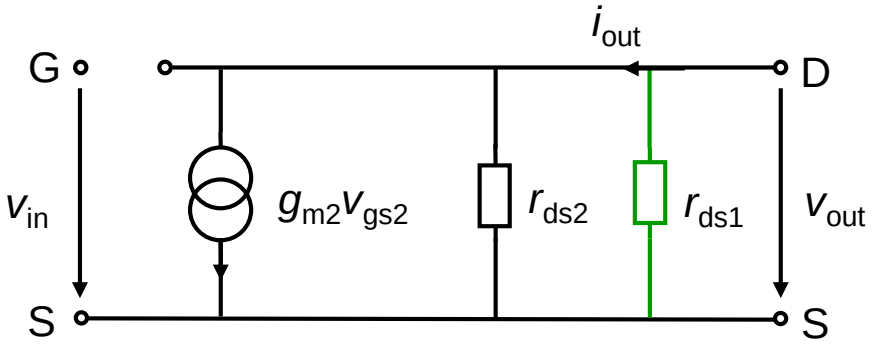


# Exercise 4 – Task 2

2.1 Draw the small signal equivalent circuit and derive an equation for the small-signal gain!




$$A_\mathrm{v} = -G_\mathrm{m} r_\mathrm{out}$$

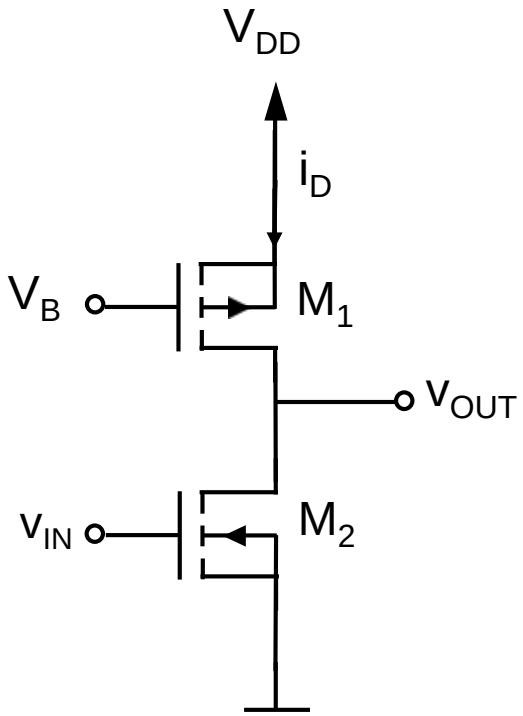
$$G_\mathrm{m} = \left.\frac{i_\mathrm{out}}{v_\mathrm{in}}\right|_{v_\mathrm{out}=0\,\mathrm{V}} = g_\mathrm{m2} = \beta_\mathrm{n}\frac{W}{L}(V_\mathrm{IN} - V_\mathrm{th2})$$

$$r_\mathrm{out} = \left.\frac{v_\mathrm{out}}{i_\mathrm{out}}\right|_{v_\mathrm{in}=0\,\mathrm{V}} = r_\mathrm{ds1} || r_\mathrm{ds2}$$

$$\Rightarrow A_\mathrm{v} = -\beta_\mathrm{n}\frac{W}{L}(V_\mathrm{IN} - V_\mathrm{th2})(r_\mathrm{ds1} || r_\mathrm{ds2})$$

# Exercise 4 – Task 2

2.2 Assuming that both transistors are operating in saturation, determine the W/L-ratios of M1 and M2




We first calculate W/L of M2 by the gain:

$$A_\mathrm{v} = -g_\mathrm{m2} r_\mathrm{out} = -72$$

$$r_\mathrm{out1,2} = \frac{1}{\lambda_\mathrm{n} I_\mathrm{D}} || \frac{1}{\lambda_p I_\mathrm{D}} = 0.2\ \mathrm{M\Omega}$$

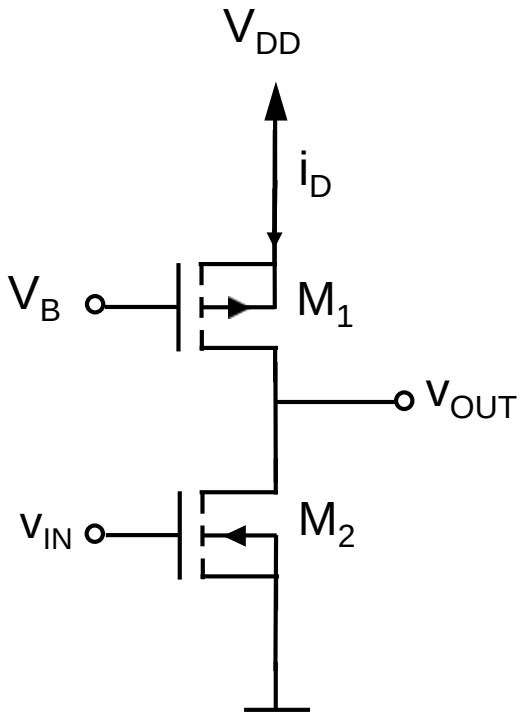
$$\Rightarrow g_\mathrm{m2} = \frac{A_\mathrm{v}}{r_\mathrm{out}} = 360\ \frac{\mu\mathrm{A}}{\mathrm{V}}$$

$$g_\mathrm{m2} = \sqrt{2\beta_\mathrm{n} I_\mathrm{D} (W/L)_2}$$

$$\Rightarrow \left(\frac{W}{L}\right)_2 = \frac{(g_\mathrm{m2})^2}{2\beta_\mathrm{n} I_\mathrm{D}} = 4$$

# Exercise 4 – Task 2

2.2 Assuming that both transistors are operating in saturation, determine the W/L-ratios of M1 and M2




Then we calculate the W/L of M1 by the maximum swing constraints:

$$v_{\mathrm{OUT,min}} = v_{\mathrm{GS2}} - \mathrm{V_{thn}} = v_{\mathrm{GS2eff}} = \sqrt{\frac{2I_{\mathrm{D}}}{\beta_{0\mathrm{n}}\left(\frac{W}{L}\right)_2}} = 0.25\ \mathrm{V}$$

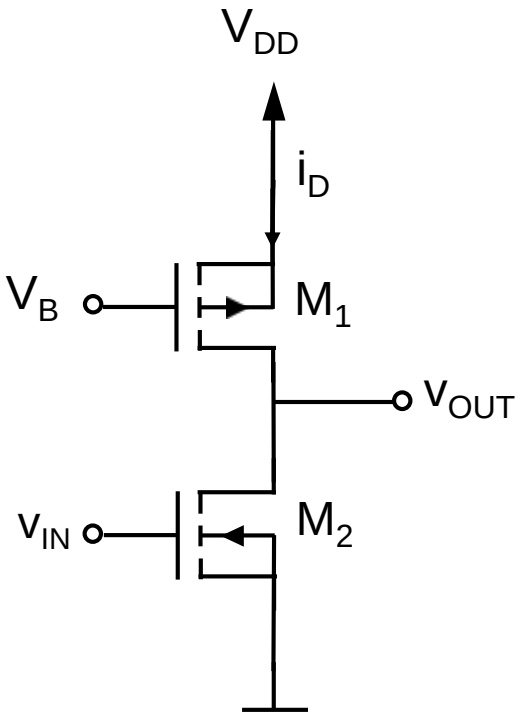
$$v_{\mathrm{OUT,max}} = v_{\mathrm{OUT,min}} + v_{\mathrm{SWING}} = 0.25\ \mathrm{V} + 0.7\ \mathrm{V} = 0.95\ \mathrm{V}$$

$$v_{\mathrm{OUT,max}} = V_{\mathrm{DD}} - v_{\mathrm{GS1eff}} = V_{\mathrm{DD}} - \sqrt{\frac{2I_{\mathrm{D}}}{\beta_{0\mathrm{p}}\left(\frac{W}{L}\right)_1}}$$

$$\Rightarrow \left(\frac{W}{L}\right)_1 = \frac{2I_{\mathrm{D}}}{\beta_{0\mathrm{p}}(V_{\mathrm{DD}} - v_{\mathrm{OUT,max}})^2} = 22.5$$

# Exercise 4 – Task 2

2.3 Determine the bias voltages $v_{IN}$ and $V_B$! Channel-length modulation is to be neglected.




$M_2$:

$$v_{\mathrm{IN}} = v_{\mathrm{GS2}} = v_{\mathrm{GS2eff}} + \mathrm{V_{thn}}$$

$$v_{\mathrm{IN}} = 0.25\ \mathrm{V} + 0.193\ \mathrm{V} = 0.343\ \mathrm{V}$$

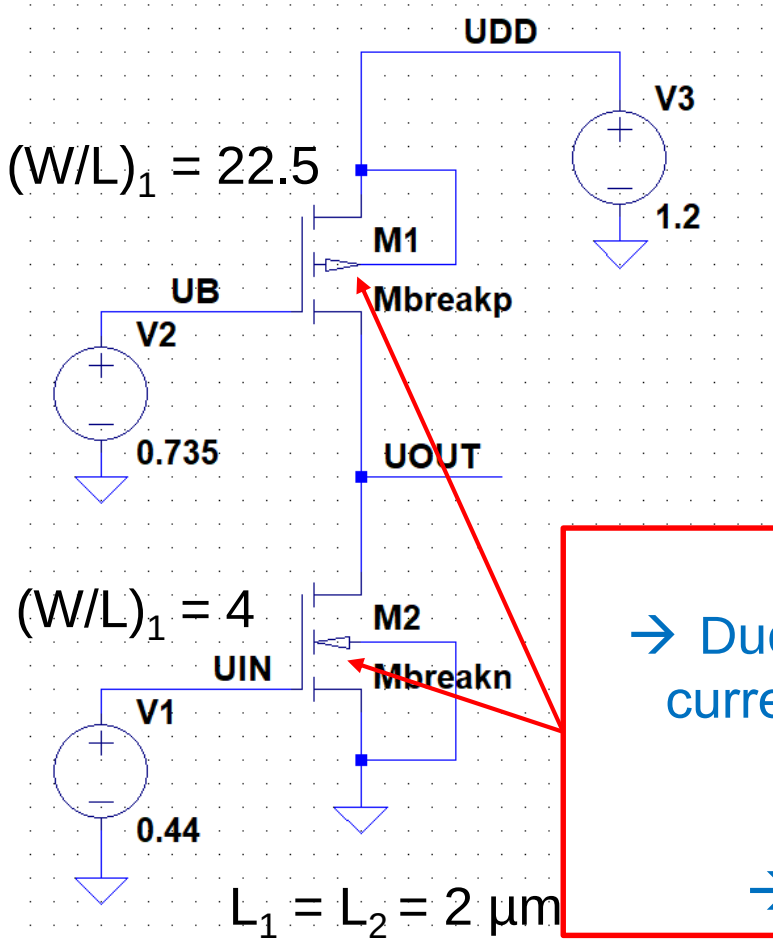
$M_1$:

$$V_{\mathrm{B}} = V_{\mathrm{DD}} - \left(v_{\mathrm{GS2eff}} + \mathrm{V_{thp}}\right)$$

$$V_{\mathrm{B}} = 1.2\ \mathrm{V} - (0.25\ \mathrm{V} + 0.215\ \mathrm{V}) = 0.735\ \mathrm{V}$$

# Exercise 4 – Task 2

2.4 Implement the circuit in LTspice. Why is a verification of the small-signal parameters and thus of the small-signal gain infeasible by means of a bias point simulation?




```
        --- Operating Point ---

V(udd):         1.2             voltage
V(ub):          0.735           voltage
V(uout):        0.993496        voltage
V(uin):         0.44            voltage
V(n002):        0.6             voltage
V(n001):        -60000          voltage
V(nc_01):       0               voltage
Id(M2):         4.48281e-005    device_current
Ig(M2):         0               device_current
Ib(M2):         -1.0035e-012    device_current
Is(M2):         -4.48281e-005   device_current
Id(M1):         4.48278e-005    device_current
Ig(M1):         0               device_current
Ib(M1):         2.16501e-013    device_current
```

Two current sources working against each other!

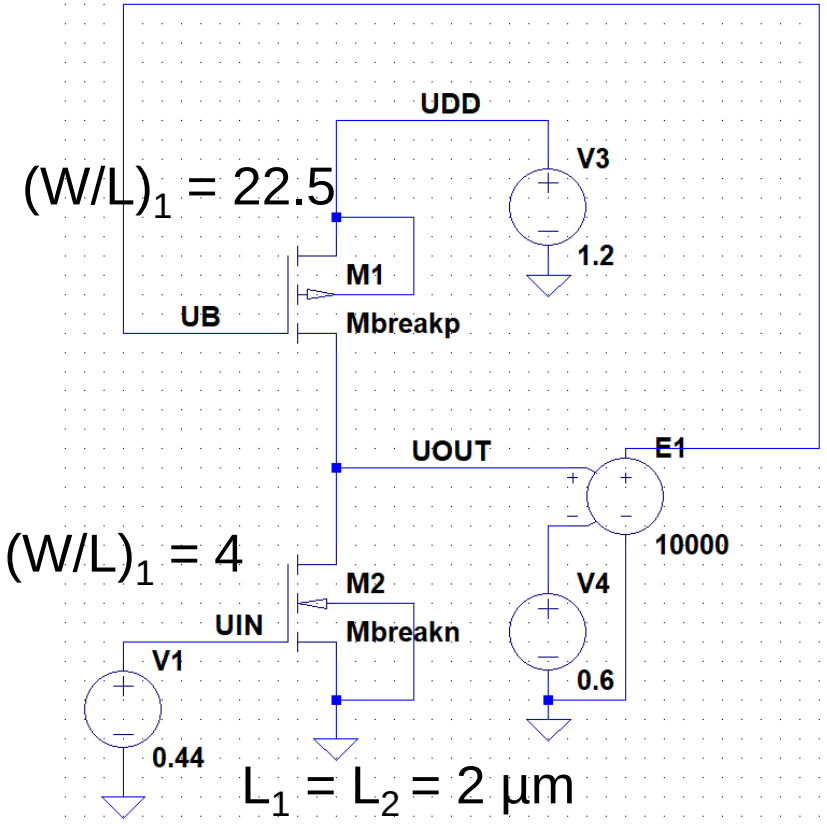
→ Due to fixed gate-source voltages: current source with higher current must "reduce" its $V_{DS}$ in order to match the lower current of the other current source.

→ $V_{out}$ close or even equal to $V_{DD}$ or $V_{SS}$.

→ One of the transistors operates in the linear region.

# Exercise 4 – Task 2

2.5 Connect $v_{OUT}$ to the non-inverting input of an ideal differential amplifier…
Perform a bias point analysis and verify your results for the small signal parameters and gain.




```
      --- Operating Point ---

V(udd):         1.2             voltage
V(ub):          0.742926        voltage
V(uout):        0.600007        voltage
V(uin):         0.44            voltage
V(nc_01):       0.735           voltage
V(n001):        0.6             voltage
Id(M2):         4.34161e-005    device_current
Ig(M2):         0               device_current
Ib(M2):         -6.10007e-013   device_current
Is(M2):         -4.34161e-005   device_current
Id(M1):         4.34161e-005    device_current
Ig(M1):         0               device_current
Ib(M1):         6.09993e-013    device_current
Is(M1):         -4.34161e-005   device_current
```

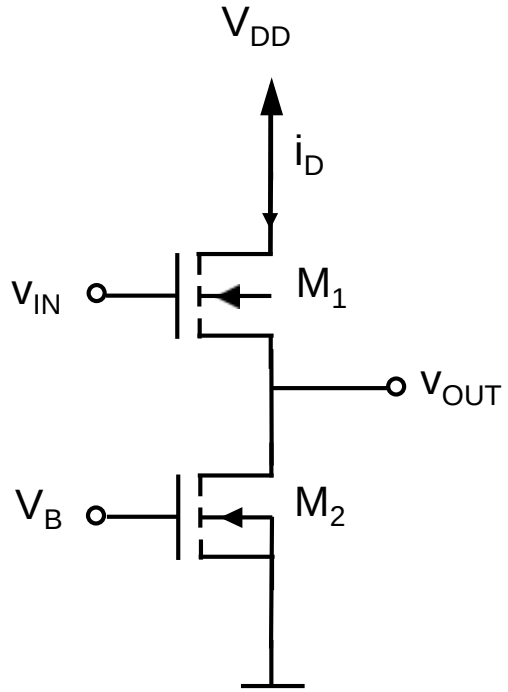
Common-mode feedback keeps $V_{OUT}$ at the optimal bias point $V_{DD}/2$!

# Exercise 4 – Task 3

## Task 3: Common-drain Amplifier (Source Follower / Level-Shifter)

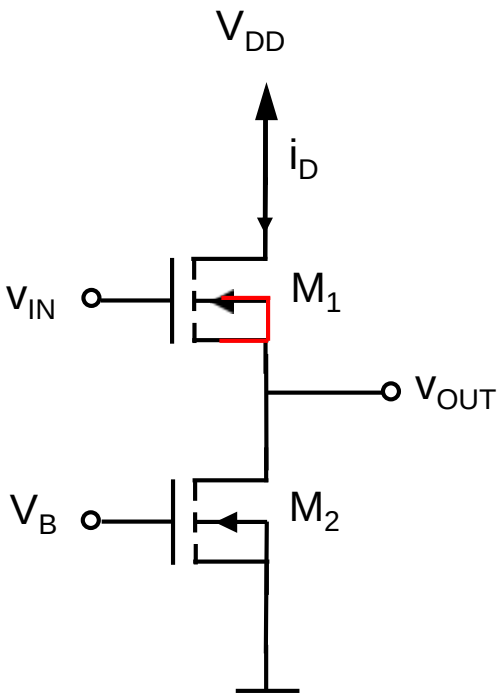
$V_{DD}$ = 1.2 V, $I_D$ = 100µA, $v_{OUT} = v_{IN}$ – 0.4 V, $v_{GS2} - v_{GS1}$ = 0.2 V

$\beta_{0n}$ = 360 µA/V$^2$, $V_{thn}$ = 0.19 V, L = 2 µm



# Exercise 4 – Task 3

3.1 Calculate the W/L-ratios of M1 and M2 for $\lambda$ = γ = 0, i.e., the channel-length modulation and the body-effect are to be neglected!




## given values

$$v_{\mathrm{OUT}} = v_{\mathrm{IN}} - 0.4\ \mathrm{V} \Rightarrow v_{\mathrm{GS1}} = v_{\mathrm{IN}} - v_{\mathrm{OUT}} = 0.4\ \mathrm{V}$$
$$v_{\mathrm{GS2}} - v_{\mathrm{GS1}} = 0.2\ \mathrm{V} \Rightarrow v_{\mathrm{GS2}} = 0.6\ \mathrm{V}$$

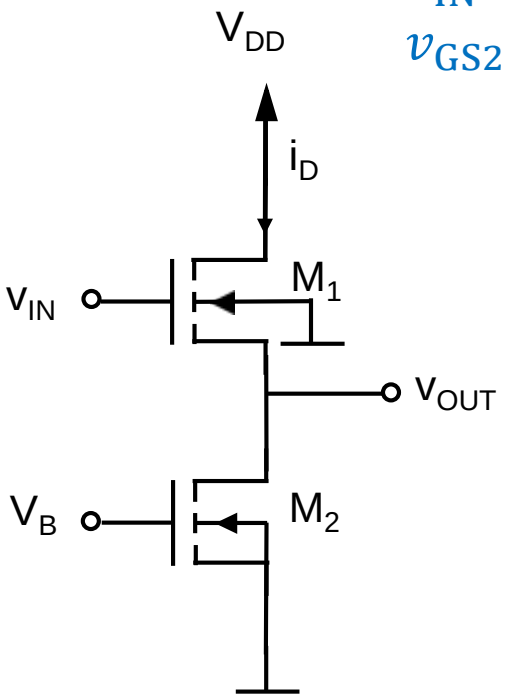
$$I_{\mathrm{D}} = 100\ \mu\mathrm{A}$$
$$V_{\mathrm{th0}} = 0.19\ \mathrm{V}$$
$$\beta_0 = 360\ \mu\mathrm{A/V}$$

## W/L-ratios without body effect

$$I_{\mathrm{D}} = \frac{\beta_0}{2}\left(\frac{W}{L}\right)_1 (V_{\mathrm{GS1}} - \mathrm{V_{th0}})^2 \Rightarrow \left(\frac{W}{L}\right)_1 = 12.6$$
$$I_{\mathrm{D}} = \frac{\beta_0}{2}\left(\frac{W}{L}\right)_2 (V_{\mathrm{GS2}} - \mathrm{V_{th0}})^2 \Rightarrow \left(\frac{W}{L}\right)_2 = 3.3$$

# Exercise 4 – Task 3

3.2 Repeat task 3.1 for $\gamma$ = 0.2 $\sqrt{V}$, $2\Phi_F$ = 0.2 $V$, and $v_{IN}$ = 1 V!




given values

$$v_{\mathrm{IN}} - v_{\mathrm{OUT}} = 0.4\ \mathrm{V} \Rightarrow v_{\mathrm{GS1}} = 0.4\ \mathrm{V}$$
$$v_{\mathrm{GS2}} - v_{\mathrm{GS1}} = 0.2\ \mathrm{V} \Rightarrow v_{\mathrm{GS2}} = 0.6\ \mathrm{V}$$

bulk effect

$$v_{\mathrm{SB1}} = v_{\mathrm{OUT}}$$
$$v_{\mathrm{IN}} = 1\ \mathrm{V}, v_{\mathrm{IN}} - v_{\mathrm{OUT}} = 0.4\ \mathrm{V} \Rightarrow v_{\mathrm{OUT}} = v_{\mathrm{SB1}} = 0.6\ \mathrm{V}$$

threshold voltage

$$V_{\mathrm{th}} = V_{\mathrm{T0}} + \gamma\left(\sqrt{V_{\mathrm{SB1}} + 2\varphi_{\mathrm{F}}} - \sqrt{2\varphi_{\mathrm{F}}}\right)$$
$$= 0.19\ \mathrm{V} + 0.2\left(\sqrt{0.6 + 0.2} - \sqrt{0.2}\right)\mathrm{V} = 0.279\ \mathrm{V}$$

W/L-ratios with bulk effect

$$I_{\mathrm{D}} = \frac{\beta_0}{2}\left(\frac{W}{L}\right)_2 (v_{\mathrm{GS2}} - V_{\mathrm{th}})^2 \Rightarrow \left(\frac{W}{L}\right)_2 = 3.3$$

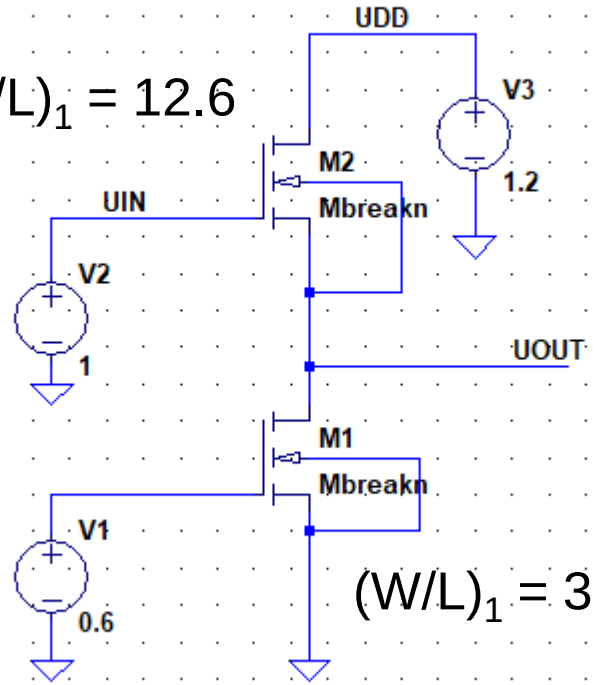
$$I_{\mathrm{D}} = \frac{\beta_0}{2}\left(\frac{W}{L}\right)_1 (v_{\mathrm{GS1}} - V_{\mathrm{th}})^2 \Rightarrow \left(\frac{W}{L}\right)_1 = 37.9$$

# Exercise 4 – Task 3

3.3 Implement the circuit in LTspice and perform a bias point simulation. Compare the results for $v_{OUT}$ and $i_D$ with the calculation.

$L_1 = L_2 = 2\ \mu m$

$(W/L)_1 = 12.6$




$(W/L)_1 = 3.3$

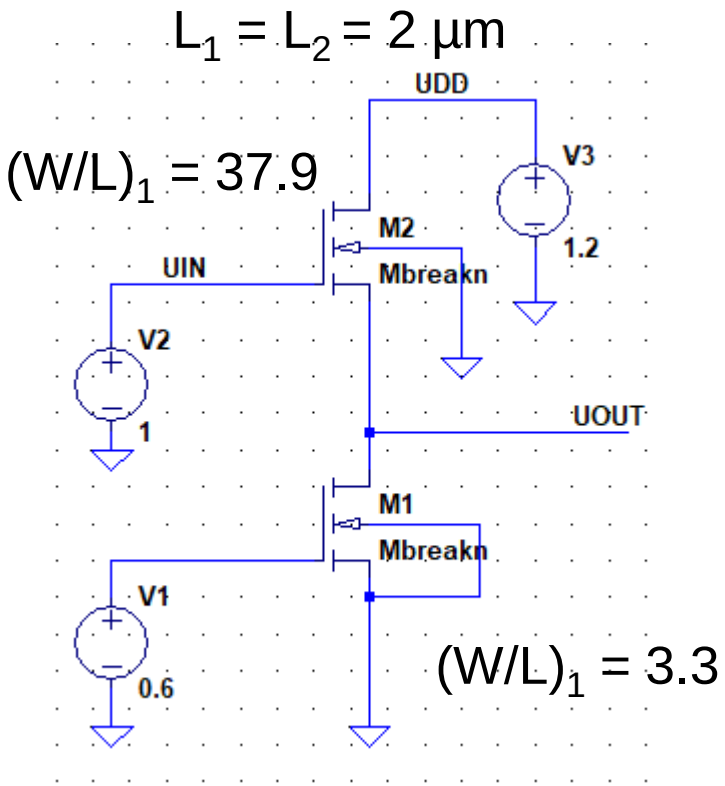
```
      --- Operating Point ---

V(n001):        0.6             voltage
V(uin):         1               voltage
V(udd):         1.2             voltage
V(uout):        0.603523        voltage
Id(M2):         9.52181e-005    device_current
Ig(M2):         0               device_current
Ib(M2):         -6.06477e-013   device_current
Is(M2):         -9.52181e-005   device_current
Id(M1):         9.52181e-005    device_current
Ig(M1):         0               device_current
Ib(M1):         -6.13523e-013   device_current
Is(M1):         -9.52181e-005   device_current
I(V3):          -9.52181e-005   device_current
I(V2):          0               device_current
I(V1):          0               device_current
```

**Simulation 1 <u>without</u> body-effect**

# Exercise 4 – Task 3

3.3 Implement the circuit in LTspice and perform a bias point simulation. Compare the results for $v_{OUT}$ and $i_D$ with the calculation.



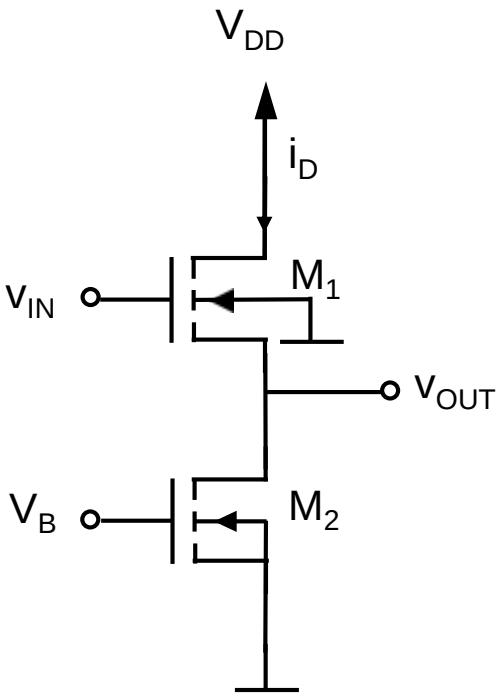

```
       --- Operating Point ---

V(n001):        0.6             voltage
V(uin):         1               voltage
V(udd):         1.2             voltage
V(uout):        0.621098        voltage
Id(M2):         9.53517e-005    device_current
Ig(M2):         0               device_current
Ib(M2):         -1.8411e-012    device_current
Is(M2):         -9.53517e-005   device_current
Id(M1):         9.53517e-005    device_current
Ig(M1):         0               device_current
Ib(M1):         -6.31098e-013   device_current
Is(M1):         -9.53517e-005   device_current
I(V3):          -9.53517e-005   device_current
I(V2):          0               device_current
I(V1):          0               device_current
```

**Simulation 2 <u>with</u> body-effect**

# Exercise 4 – Task 3

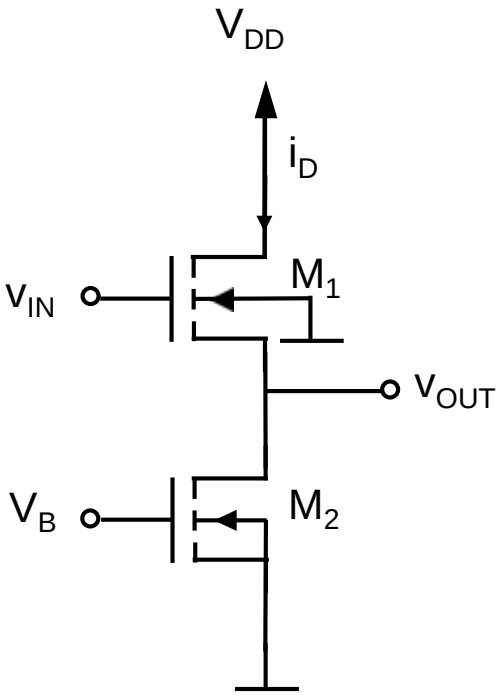
3.3 Implement the circuit in LTspice and perform a bias point simulation. Compare the results for $v_{OUT}$ and $i_D$ with the calculation.




| | **calculation** | **w/o body-effect** | **w. body-effect** |
|---|---|---|---|
| $v_{OUT}$ | 600 mV | 603 mV | 621 mV |
| $i_D$ | 100 µA | 95.2 µA | 95.3 µA |

# Exercise 4 – Task 3

3.4 Considering task 3.2, what is the minimum input voltage for which $M_2$ operates in saturation?




saturation condition for $M_2$

$$v_{\mathrm{DS2sat}} = v_{\mathrm{GS2}} - V_{\mathrm{T2}} = 0.6\,V - 0.2\,V = 0.4\,V$$

$$v_{\mathrm{SB1}} = v_{\mathrm{OUT}} = v_{\mathrm{DS2sat}}$$

$$v_{\mathrm{SB1}} = 0.6\ \mathrm{V} - 0.2\,V = 0.4\ \mathrm{V}$$

threshold voltage

$$V_{\mathrm{th}} = V_{\mathrm{T0}} + \gamma\left(\sqrt{v_{\mathrm{SB1}} + 2\varphi_{\mathrm{F}}} - \sqrt{2\varphi_{\mathrm{F}}}\right)$$

$$= 0.2\ \mathrm{V} + 0.2\left(\sqrt{0.4 + 0.2} - \sqrt{0.2}\right)\mathrm{V} = 0.265\ \mathrm{V}$$

gate-source voltage

$$I_{\mathrm{D}} = \frac{\beta_0}{2}\left(\frac{W}{L}\right)_1 (v_{\mathrm{GS1}} - V_{\mathrm{th}})^2 \Rightarrow v_{\mathrm{GS1}} = 0.376\ \mathrm{V}$$

minimum input voltage

$$v_{\mathrm{GS1}} = v_{\mathrm{IN}} - v_{\mathrm{OUT}} \Rightarrow v_{\mathrm{IN,min}} = v_{\mathrm{GS1}} + v_{\mathrm{DS2,sat}} = 0.776\ \mathrm{V}$$